

AI-Powered Phishing Detection System

Problem Statement Document — Phase 1, Week 1

Project Area: Artificial Intelligence / Machine Learning / Cybersecurity

Primary Focus: Machine-learning-based phishing URL detection

1. Background

Phishing is a major cybersecurity threat in which attackers impersonate trusted entities or services to deceive users into interacting with malicious websites, links, or other resources. The objective may include stealing credentials, financial information, personal data, or other sensitive information. IEEE research describes phishing as a continuing and serious cybercrime problem and identifies automated detection as an important defensive requirement.

URL-based detection is an important research direction because a URL contains structural and lexical information that can be extracted without requiring the detection system to automatically visit the suspicious website. Existing IEEE research has investigated URL features with traditional machine-learning models, ensemble learning, deep learning, browser-side detection, and early detection of newly registered or parked domains.

2. Problem Context

Traditional phishing protection mechanisms commonly include blacklists, whitelists, heuristic rules, and manually maintained threat intelligence. These mechanisms are useful, but they can struggle when a malicious URL is newly created, modified, obfuscated, or not yet present in a known threat database. Research has therefore explored machine-learning methods that learn patterns from phishing and legitimate URLs.

However, a machine-learning detector introduces its own challenges. A model can perform very well on the dataset used for training and testing while performing less effectively on URLs from a different source. Recent IEEE Access research on XGBoost-based phishing detection specifically demonstrates the importance of cross-dataset validation and generalization.

3. Core Problem Statement

Existing phishing URL detection approaches can achieve high detection performance under controlled experimental conditions, but their effectiveness may be limited by dataset dependency, changing phishing patterns, feature selection, false positives, and reduced generalization to unseen or independently collected URLs. Therefore, there is a need for an AI-powered phishing detection system that systematically extracts meaningful URL characteristics, compares machine-learning approaches, evaluates their performance using appropriate metrics, and investigates whether the resulting model generalizes beyond the dataset on which it was developed.

4. Research Problem

The research problem is not simply to classify a URL as phishing or legitimate. The project must investigate whether machine-learning models can provide reliable URL-based phishing detection while maintaining acceptable performance on previously unseen data.

Challenge	Research concern
Dataset dependency	A model may learn characteristics specific to one dataset rather than general phishing patterns.
Generalization	Performance can decrease when the model is evaluated on an independent dataset.
Feature selection	Large URL feature sets may contain redundant or unstable features.

Challenge	Research concern
Evolving attacks	Attackers continuously create and modify phishing URLs.
False positives and false negatives	Incorrect classification can either expose users to threats or incorrectly block legitimate resources.
Explainability	Users and researchers benefit from understanding which URL characteristics influence a prediction.

5. Proposed Problem-Solving Direction

The proposed project will investigate a URL-based AI/ML pipeline in which a submitted URL is validated, transformed into measurable lexical and structural features, and passed to trained classification models. The system will produce a classification and a phishing-risk probability. Multiple algorithms can be compared, with XGBoost considered as a strong candidate for the primary model because of its suitability for structured/tabular URL features.

Proposed high-level workflow:

User URL → URL Validation → Feature Extraction → Feature Selection/Preprocessing → ML Classifier → Phishing Probability → Risk Classification → Result/Explanation

6. Project Objectives

1. Study existing phishing URL detection approaches through the selected research literature.
2. Identify important URL characteristics that can support phishing classification.
3. Prepare and preprocess suitable phishing and legitimate URL datasets.
4. Implement and compare multiple machine-learning classification algorithms.
5. Evaluate models using accuracy, precision, recall, F1-score, specificity, ROC-AUC and confusion matrices where appropriate.
6. Investigate model generalization using data that is independent from the training dataset.
7. Provide a usable interface through which a user can submit a URL for analysis.
8. Provide an interpretable result containing classification, probability/risk information and relevant feature information where feasible.

7. Scope of the Problem

In scope: URL validation, URL feature extraction, preprocessing, feature selection, supervised machine learning, model comparison, evaluation, cross-dataset validation, prediction probability, risk classification, and a web-based demonstration interface.

Initially out of scope: automatic browsing of suspicious websites, credential collection, automatic attack execution, full browser/network interception, and guaranteed detection of every phishing attack. These may be considered future extensions.

Reference note: The problem formulation is intentionally broader than a single algorithm. Students should use the literature to justify their eventual choice of dataset, features, models and evaluation methodology.